

Rupali Bhati

EDUCATION

Northeastern University (Supervisor: Christopher Amato)
Ph.D., Computer Science
GPA: 3.92/4.0

Boston, U.S.A.
Sep 2023 - Present

Université Laval/ Mila (Supervisor: Audrey Durand)
Masters, Computer Science (with thesis)
GPA: 4.2/4.3

Quebec, Canada
Sep 2020 - Aug 2023

Delhi Technological University (Supervisor: Indu Sreedevi)
Bachelors, Electronics and Communication Engineering
Aggregate percentage: 72.29% (WES equivalent 3.55/4.0)

New Delhi, India
Aug 2012 - May 2016

PUBLICATIONS

12. Decentralized Asymmetric DQN: Decentralization without Factorization in Multi-Agent Reinforcement Learning [\[link\]](#)
Rupali Bhati*, Anurag Kadkol*, Andrea Baisero, Christopher Amato (*= equal contribution)
RLC 2026

11. Fixing Incomplete Value Function Decomposition for Multi-Agent Reinforcement Learning [\[link\]](#)
Andrea Baisero, **Rupali Bhati**, Shuo Liu, Aathira Sunil Pillai, Christopher Amato
RLC 2026

10. The Influence of Scaffolds on Coordination Scaling Laws in LLM Agents [\[link\]](#)
Rupali Bhati*, Mariana Meireles*, Niklas Lauffer[†], Cameron Allen[†] (*= equal contribution, [†] equal advising)
NeurIPS 2025 Scaling Environments for Agents Workshop

9. On Stateful Value Factorization in Multi-Agent Reinforcement Learning [\[link\]](#)
Enrico Marchesini, Andrea Baisero, **Rupali Bhati**, Christopher Amato
AAMAS 2025

8. Use of an Integrated Knowledge Translation Approach to Develop an Electronic Patient-Reported Outcome System for Cancer Rehabilitation: Tutorial [\[link\]](#)
Christian Lopez, Sarah E Neil-Sztramko, Kristin L Campbell, David M Langelier, Tran Truong, Yuliya Gavrylyuk, Pia Nyakairu, Laura Parente, Audrey Durand, Jackie L Bender, Gillian Strudwick, **Rupali Bhati**, Jonathan Greenland, Tony Reiman, Jennifer M Jones
JMIR Cancer 2025

7. Scalable Approaches for a Theory of Many Minds [\[link\]](#)
Maximilian Puelma Touzel, Amin Memarian, Matthew Riemer, Andrei Mircea Romascanu, Andrew Williams, Elin Ahlstrand, Lucas Lehnert, **Rupali Bhati**, Guillaume Dumas, Irina Rish
ICML 2024 Agentic Markets Workshop

6. Curriculum Learning for Cooperation in Multi-Agent Reinforcement Learning [\[link\]](#)
Rupali Bhati, SaiKrishna Gottipati, Clodéric Mars, Matthew E. Taylor
NeurIPS 2023 Agent Learning in Open-Endedness Workshop

5. Performative Prediction in Time Series: A Case Study [\[link\]](#)
Rupali Bhati, Jennifer Jones, Kristin Campbell, David Langelier, Anthony Reiman, Jonathan Greenland, Audrey Durand
NeurIPS 2022 Workshop on Learning from Time Series for Health

4. Summarizing Societies: Agent Abstraction in Multi-Agent Reinforcement Learning [\[link\]](#)
Amin Memarian, Maximilian Puelma Touzel, Matthew D Riemer, **Rupali Bhati**, Irina Rish
ICLR 2022 From Cells to Societies: Collective Learning across Scales Workshop

3. Interpret Your Care: Predicting the Evolution of Symptoms for Cancer Patients [\[link\]](#)

Rupali Bhati, Jennifer Jones, Audrey Durand

AAAI 2022 Trustworthy AI for Healthcare Workshop

2. CARL: Conditional-value-at-risk Adversarial Reinforcement Learning [\[link\]](#)

Mathieu Godbout, Maxime Heuillet, Sharath Chandra, **Rupali Bhati**, Audrey Durand

AAAI 2022 Safe AI Workshop

1. A Reinforcement Learning Approach to Jointly Adapt Vehicular Communications and Planning for Optimized Driving [\[link\]](#)

Mayank K. Pal, **Rupali Bhati**, Anil Sharma, Sanjit K. Kaul, Saket Anand, P.B.Sujit

IEEE ITSC 2018

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Intern, Center for Human Compatible AI (CHAI), UC Berkeley

Supervisor: Niklas Lauffer

May 2025 - Aug 2025

- Worked on a project exploring the effect of scaffolding on cooperative strategies between LLMs of different skill levels.

Graduate Research Assistant, Northeastern University

Supervisor: Christopher Amato

Sep 2023 - Present

- Currently working on implementing the Box pushing environment in Jax to include as a part of the JAXMarl suite of environments.
- Currently working on a project on Decentralised asymmetric DQN. Testing the method on the JAXMarl suite of environments.
- Worked on fixing incomplete value function decomposition for multi-agent reinforcement learning.
- Worked on stateful value factorization in multi-agent reinforcement learning.

Machine Learning Alignment & Theory (MATS) Scholar

Supervisor: Christian Schroeder de Witt

June 2024 - Aug 2024

- Explored the validity of the individual global max (IGM) principle during training in value decomposition methods using multi-agent reinforcement learning.

Research Intern

AI Redefined

Jan 2023 - Jul 2023

- Worked on achieving cooperation in multi-agent settings via curriculum learning and reinforcement learning in the game of Overcooked.

Graduate Research Assistant, Université Laval & Mila

Supervisor: Audrey Durand

Sep 2020 - Aug 2023

- Addressed the problem of performative prediction in time-series data for predicting cancer-related fatigue and pain and successfully found stable points by applying repeated performative training.
- Formulated agent abstraction in the multi-agent setting and showed how it can help disentangle non-stationarity in the game of Diplomacy and achieve higher compression.

Reinforcement Learning Consultant

Self-Employed

Sep 2018 - Aug 2020

- At Bert Labs, applied RL to increase the energy efficiency of a HVAC system. For a leading global FMCG company's headquarters building, using DQN, increased efficiency of their Air-Handling Unit system by over 70% as compared to classical PID logic.
- Conducted a week long workshop to teach fundamentals of RL to employees at Adventum. Consulted on application of RL to improve segmentation in medical images.
- Worked with CatapulZ to develop RL blue agents to Capture-The-Flag in cybersecurity applications.
- Worked at UpGrad as a Domain Expert to develop an end-to-end solution for a model inventory management problem to meet next-to-next day demand using DDQN.

Research Assistant, Indraprastha Institute of Information Technology - Delhi

Supervisor: Saket Anand

Sep 2017 - Aug 2018

- Trained an autonomous vehicle to smartly adapt communications and planning actions, while achieving large driving utilities using Q-learning.

Data Analyst

KPMG

Jun 2016 - Aug 2017

- In collaboration with Microsoft, developed an algorithm using policy iteration for automating 'Dynamic Pricing of Tickets' to maximise revenue and help reduce human effort by upto 70-80%.
- Researched use cases of predictive and descriptive analytics to provide business insights to various government organisations which helped them automate processes and boost efficiency.

SCHOLARSHIPS AND AWARDS

- 2023 Khoury Distinguished Fellowship.
- 2023 Awarded Sony Interactive Entertainment Scholarship to attend the Summer School on AI and Games. Awarded first place at Game AI Jam at the Summer School on AI and Games.
- 2022 **Google CSRMP**: Selected for Google Computer Science Research Mentorship Program with mentor Wenhao Yu.
- 2022 Second place at the Rendez-Vous IA Quebec.
- 2022 Institute of Intelligence and Data (IID) Laval Tuition Scholarship.
- 2022 Nominated for **Women in Artificial Intelligence Awards** North America.

TEACHING EXPERIENCE

- **Teaching Assistant**, *CS 5100: Foundations of AI*, Northeastern University Fall 2025
- **Teaching Assistant**, *GIF-7005: Introduction to ML*, Université Laval Fall 2021
- **Mentor**, Codementor [\[link\]](#) Fall 2019 - Summer 2020
- **Teaching Assistant**, Coding Blocks [\[link\]](#) Summer 2018
- **Teaching Assistant**, UpGrad Fall 2018

SERVICE

Organiser: Workshop on Multi-Agent Learning and Its Opportunities in the Era of Generative AI at ICLR 2025 [\[link\]](#)

Organiser: Coordination and Cooperation in Multi-Agent Reinforcement Learning (CoCo-MARL) Workshop at The Reinforcement Learning Conference 2024 [\[link\]](#)

Organiser: Multi-Agent Learning Seminar [\[link\]](#)

Reviewer: Cooperative AI Summer School applications 2025, WiML Workshop at NeurIPS 2025, NeurIPS 2023, Montreal AI Symposium 2022, ITSC 2018

Facilitator: ICML WiML UnWorkshop: Machine Learning for Physical Sciences 2022
